



CODEIGNITER CHEAT SHEET

Core setup, initialization, operations & integration quick reference

PART 1: SETUP & CORE

01. SETUP & CONFIG

Installation

```
composer create-project codeigniter4/appstart...
php spark serve
.env: CI_ENVIRONMENT = development
.env: database.default.hostname = localhost
app/Config/Routes.php for routing configurati...
```

04. MODELS

Workflow

```
class UserModel extends Model {
    protected $table = 'users';
    protected $allowedFields = ['name', 'email'];
    protected $useTimestamps = true;
    protected $validationRules = ['email' => 'req...
}
$user = $model->find(1);
$model->save(['name'=>'Alice', 'email'=>'a@b.c...
$model->where('active', 1)->findAll();
```

02. ROUTING

Architecture

```
$routes->get('/users', 'UserController::index...
$routes->post('/users', 'UserController::crea...
$routes->resource('users'); // RESTful resou...
$routes->get('/users/{:num}', 'UserController...
$routes->group('api', function($routes) { .....
```

05. VIEWS

CRUD API

```
// In controller: return view('users/index', ...
<?= esc($user->name) ?>
<?php foreach ($users as $user): ?>
<li><?= esc($user->name) ?></li>
<?php endforeach ?>
<?= view('partials/header') ?>
<?= form_open('users/create') ?>
<?= form_input('name', '', ['class'=>'form-co...
```

03. CONTROLLERS

YAML / Config

```
class UserController extends BaseController {
    public function index(): string {
        $model = new UserModel();
        $data = ['users' => $model->findAll()];
        return view('users/index', $data);
    }
    public function show(int $id): ResponseInterf...
    return $this->response->setJSON(model(UserMod...
    }
}
```

06. QUERY BUILDER

Integration

```
$builder = $db->table('users');
$query = $builder->select('name,email')
->where('active', 1)
->orderBy('name')
->get(10, 0); // limit 10, offset 0
$results = $query->getResultArray();
$builder->insert(['name'=>'Alice', 'email'=>'a...
$db->transStart(); ... $db->transComplete();
```



SECURITY, MONITORING & OPERATIONS

Production hardening, observability, performance tuning & CLI reference

PART 2: OPS & SECURITY

07. FORM VALIDATION

Hardening

```
$rules = [
    'name' => 'required|min_length[2]|max_length[...
    'email' => 'required|valid_email|is_unique[us...
];
if (!$this->validate($rules)) {
    return redirect()->back()->withInput()
->with('errors', $this->validator->getErrors(...
}
```

10. COMMON HELPERS

Unit Tests

```
url_title($str) // 'My Article' 'my-articl...
word_wrap($str, 76)
excerpt($str, $phrase, $radius)
character_limiter($str, 100)
number_to_currency($n, 'USD')
php spark make:controller/model/migration/ent...
```

08. LIBRARIES & HELPERS

Observability

```
$email = \Config\Services::email();
$email->setTo('a@b.com'); $email->send();
// Load helper functions:
helper(['url', 'form', 'text']);
base_url('assets/css/main.css')
site_url('users') // includes index.php if n...
```

11. COMMON GOTCHAS

Commands

\$allowedFields prevents mass assignment vulnerabilities
esc() everything from user input prevent XSS
CSRF protection enabled by default for POST requests
Remove index.php from URL: configure RewriteBase in .htaccess
Config files in app/Config don't edit vendor files

09. SESSIONS

Optimization

```
$session = session();
$session->set('user_id', $user->id);
$session->get('user_id');
$session->remove('user_id');
$session->destroy();
// Flash data (one-time):
$session->setFlashdata('success', 'Saved!');
$session->getFlashdata('success');
```